MLOPS Assignment-2

Tracking a Pre-trained Sentiment Model with MLflow

1.Which run had the lowest latency, and why might that be?

run_batch4 had the lowest latency of 0.049 seconds.run_batch8 took 0.054 seconds.run_batch1 was the slowest, taking 0.135 seconds.Batch size 4 provided a good balance between processing multiple sentences together and reducing execution overhead. Small variations in execution time may also occur due to system performance.

2. Did accuracy change between runs? Why or why not?

No, the accuracy remained the same at 1.0 (100%) for all three runs.

- All three batch sizes (1, 4, and 8) produced the same predictions.
- Batch size changes how many sentences are processed together, not the model's prediction logic.
- Therefore, the accuracy remained unchanged while the execution time varied.

3. Name two benefits of using MLflow here instead of printing results in the terminal.

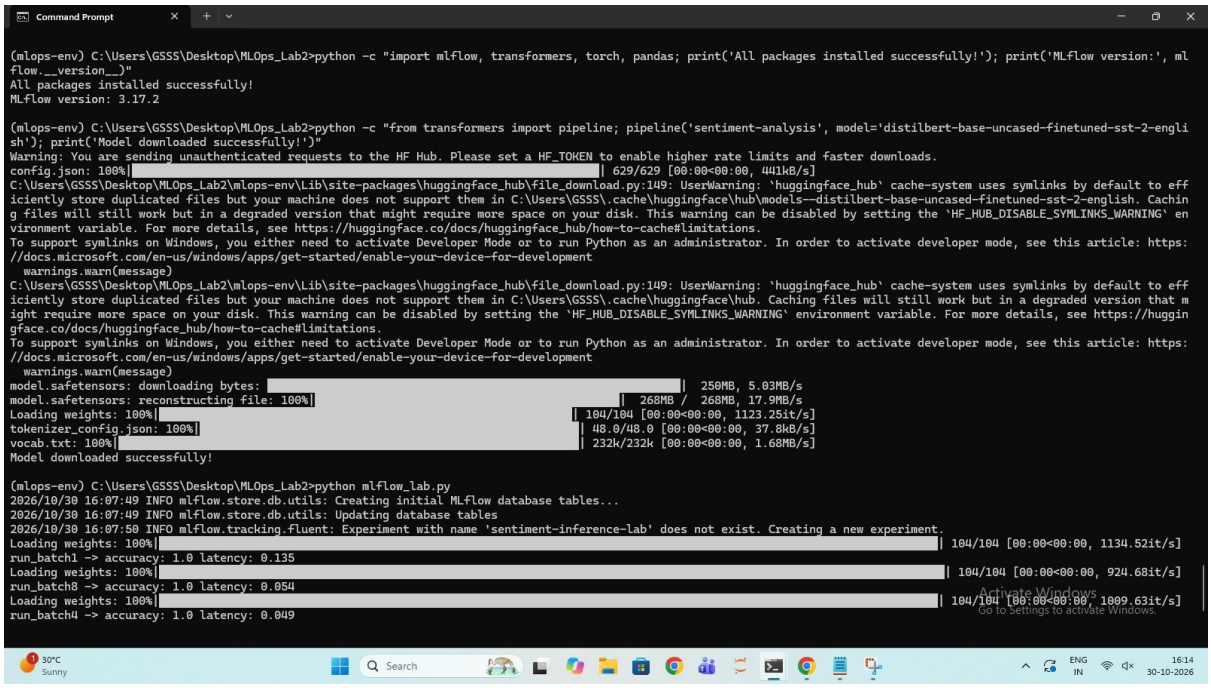
1. Experiment Tracking: MLflow automatically stores parameters such as batch size, model name, and performance metrics such as accuracy and latency. This makes it easy to track and compare different experiments.

2. Visualization and Comparison: MLflow provides a graphical dashboard with tables, charts, and comparison tools. It helps identify the best-performing run without manually checking terminal outputs.

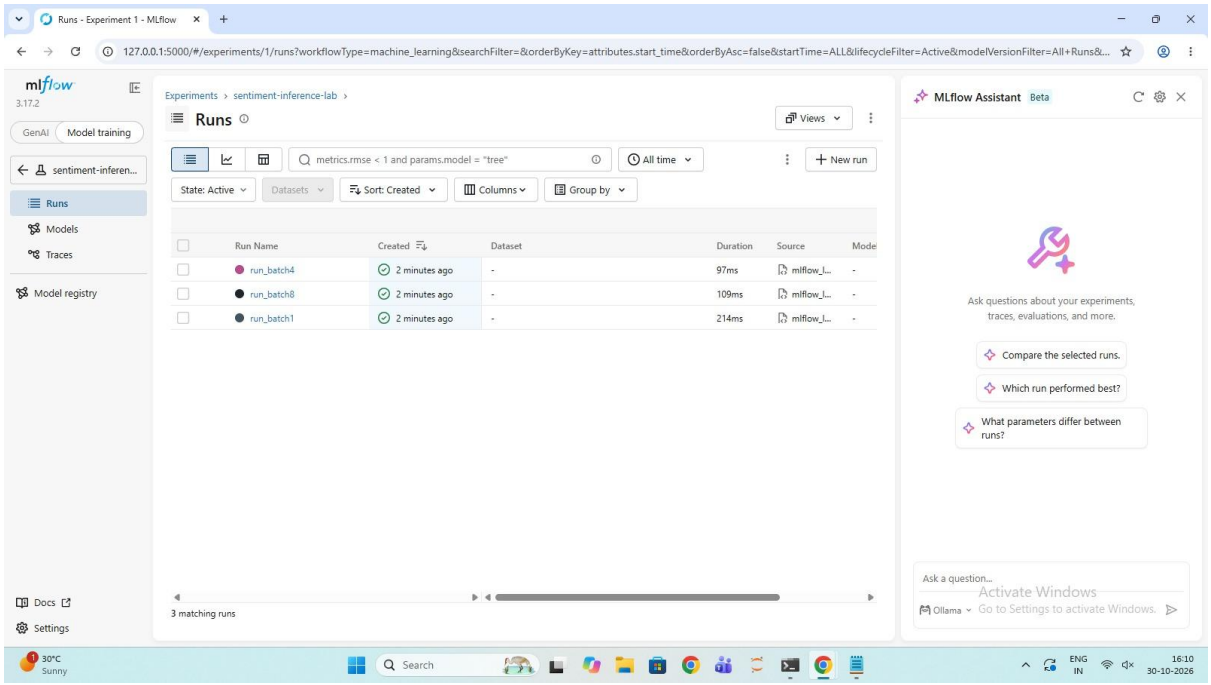
TOOLS AND TECHNOLOGIES USED:

- Windows 11
- Python 3.12
- Visual Studio Code
- MLflow 3.17.2
- Hugging Face Transformers

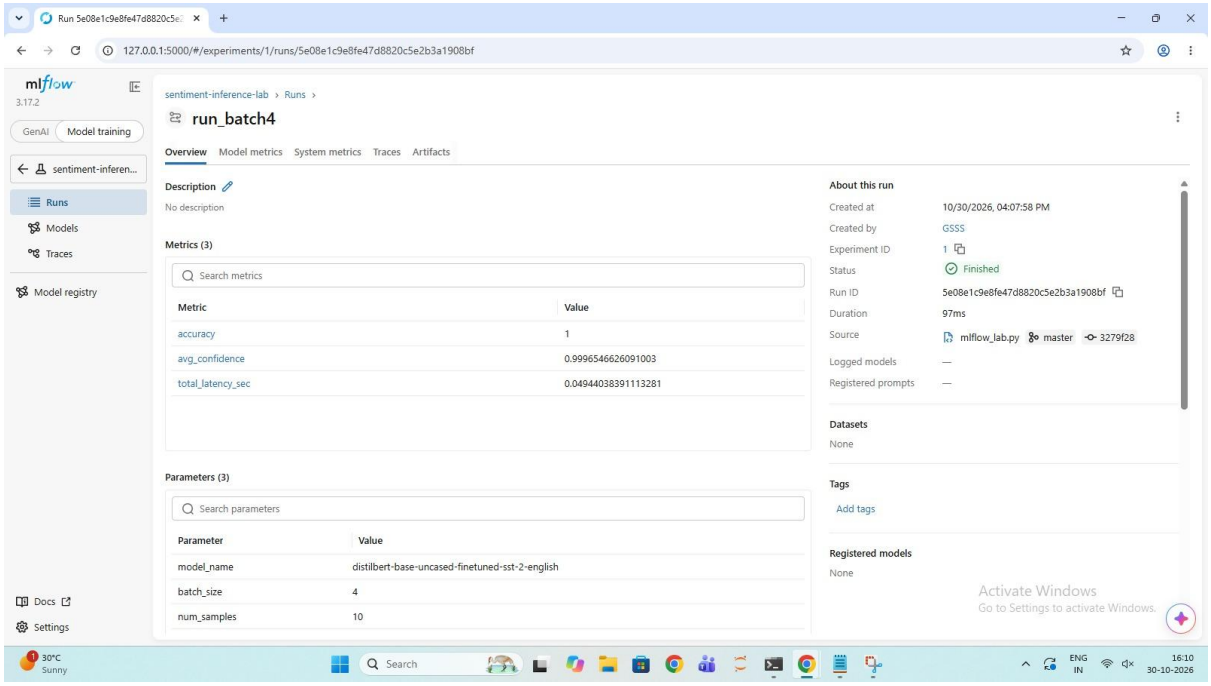
- PyTorch (CPU)
- Pandas
- DistilBERT Pre-trained Model
- Python Virtual Environment (venv)
- Command Prompt (CMD)
- MLflow Web UI (localhost:5000)



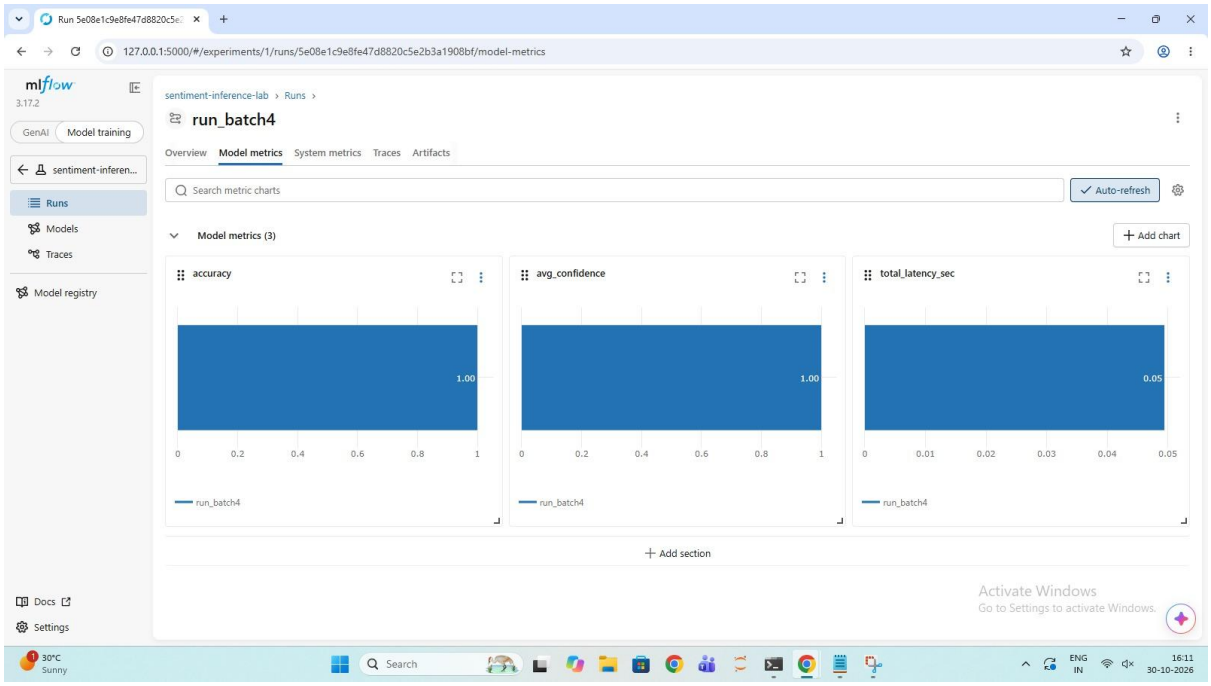
MLflow Experiment Dashboard Showing All Three Runs:



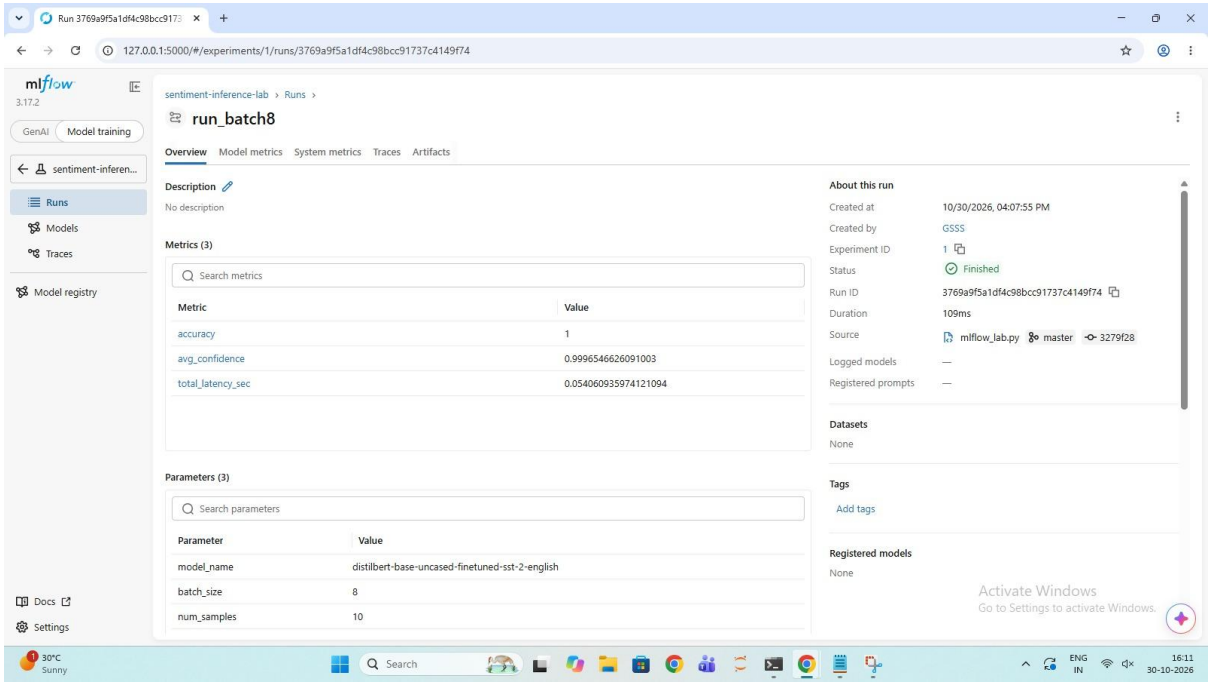
Overview of MLflow Run – Batch Size 4:



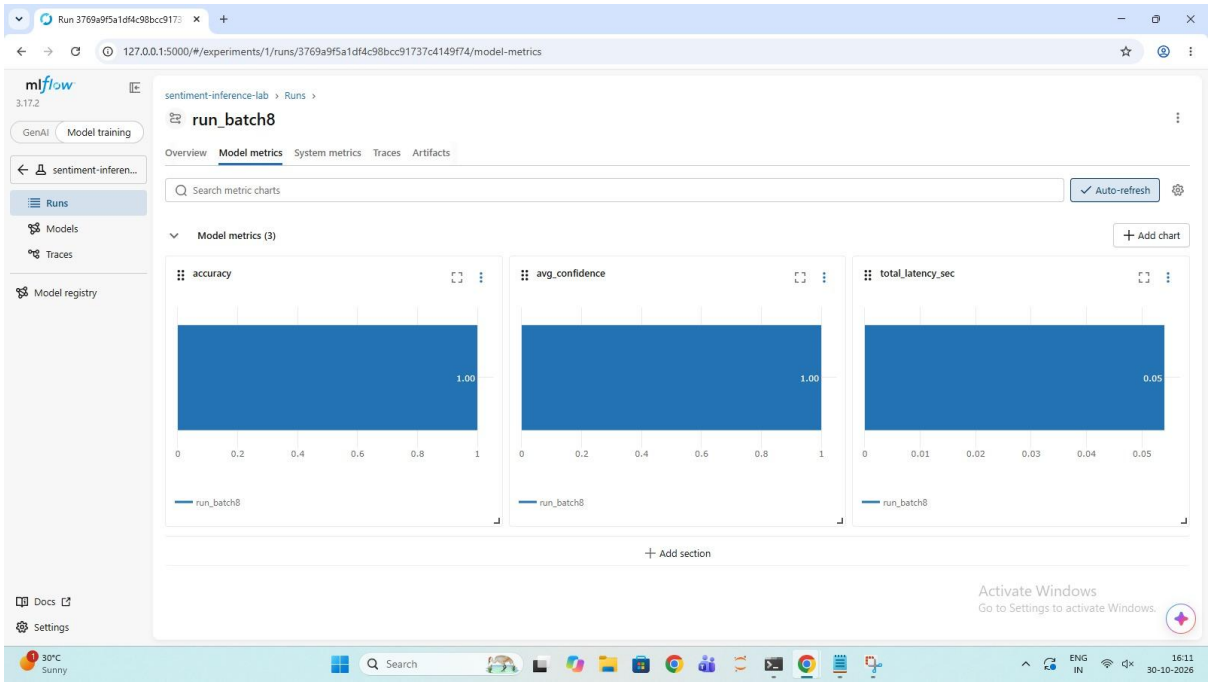
Performance Metrics and Parameters of Batch Size 4:



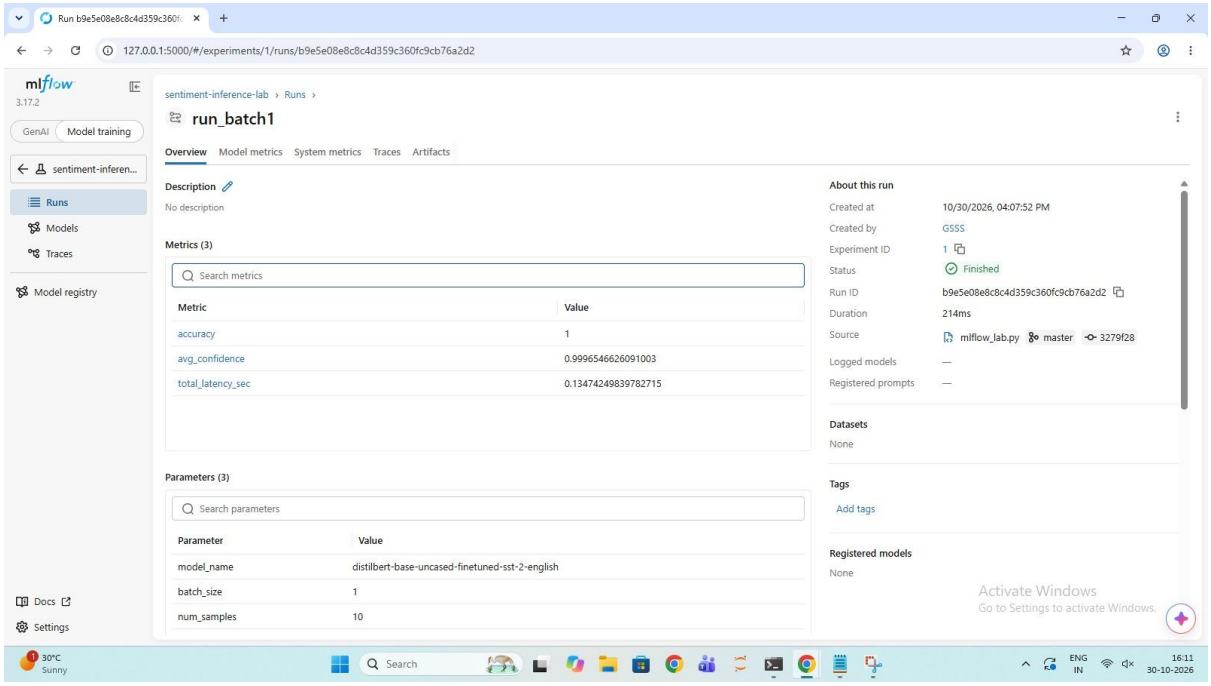
Overview of MLflow Run – Batch Size 8:



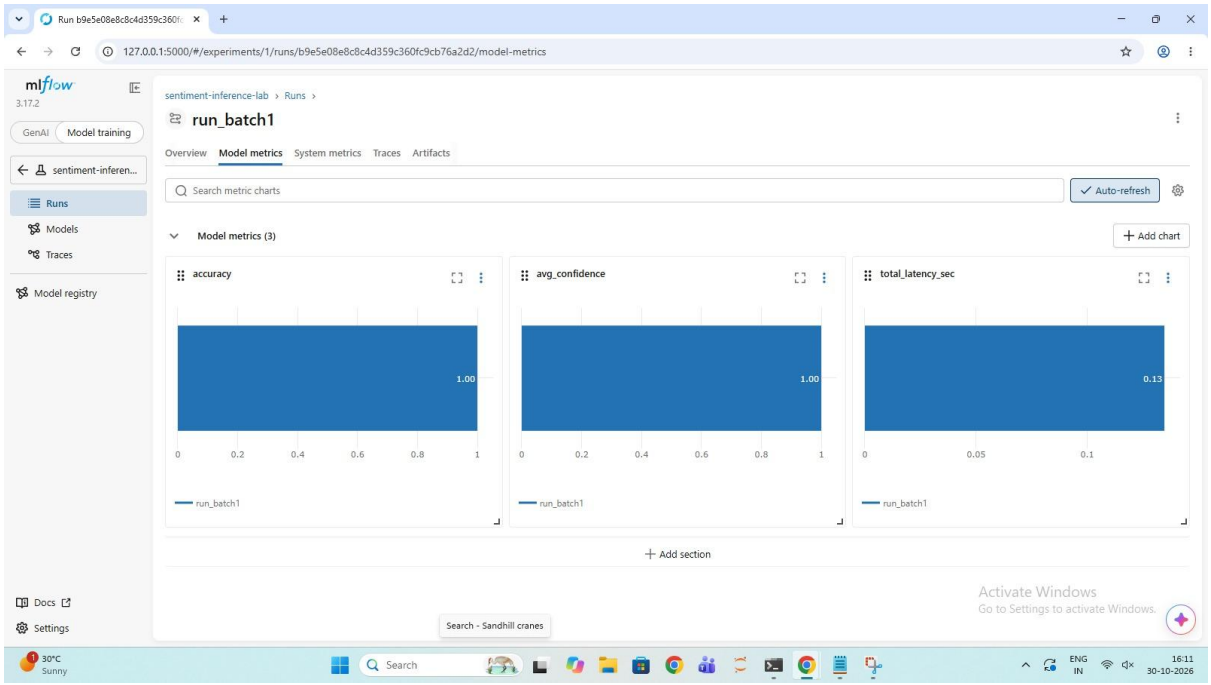
Model Performance Metrics for Batch Size 8:



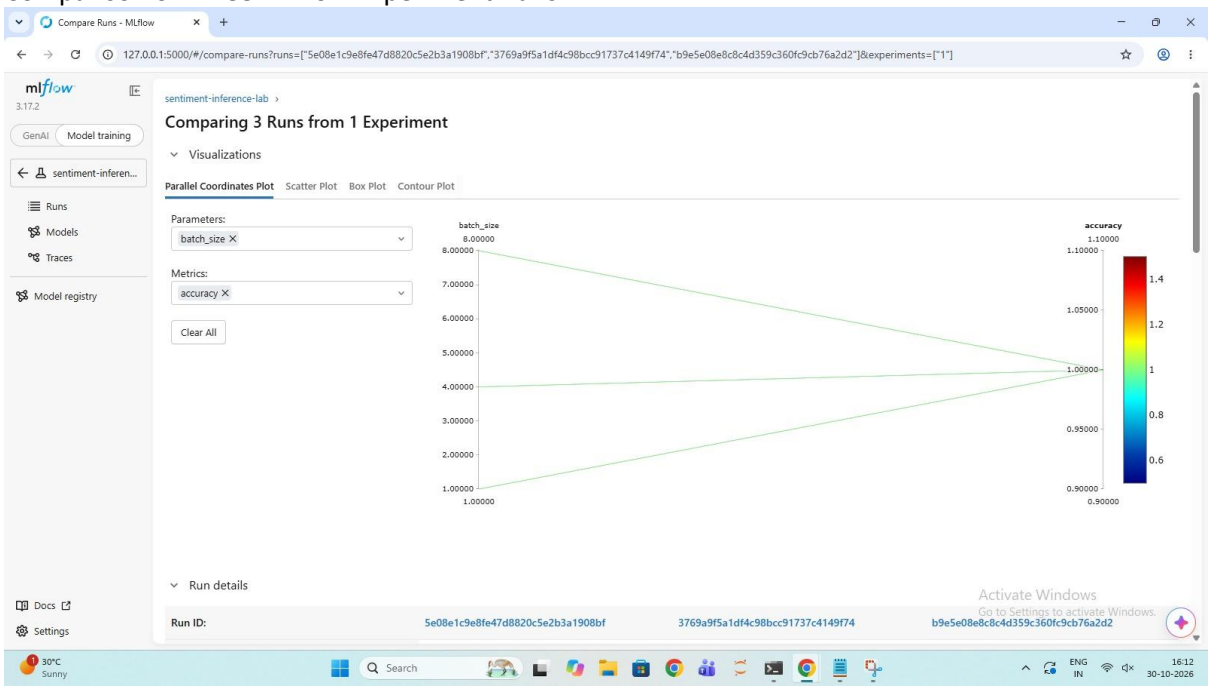
Overview of MLflow Run – Batch Size 1:



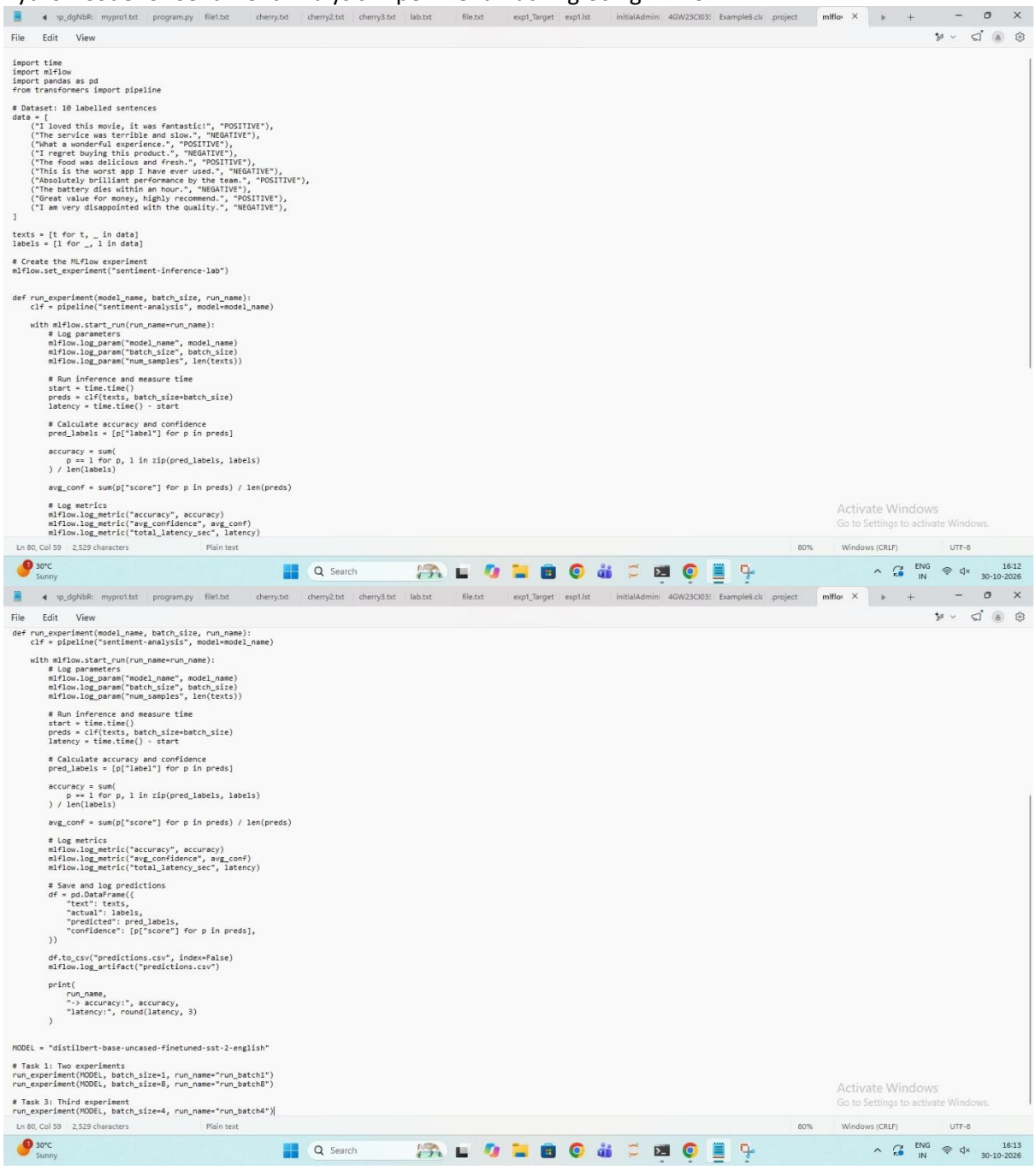
Model Performance Metrics for Batch Size 1:



Comparison of Three MLflow Experiment Runs:



Python Code for Sentiment Analysis Experiment Tracking Using MLflow:



```
import time
import mlflow
import pandas as pd
from transformers import pipeline

# Dataset: 10 labelled sentences
data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]

texts = [t for t, _ in data]
labels = [1 for _, l in data]

# Create the MLflow experiment
mlflow.set_experiment("sentiment-inference-lab")


def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)

    with mlflow.start_run(run_name=run_name):
        # Log parameters
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        # Run inference and measure time
        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

        # Calculate accuracy and confidence
        pred_labels = [p["label"] for p in preds]

        accuracy = sum(
            p == l for p, l in zip(pred_labels, labels)
        ) / len(labels)

        avg_conf = sum(p["score"] for p in preds) / len(preds)

        # Log metrics
        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)
```

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```
def run_experiment(model_name, batch_size, run_name):
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        ) / len(labels)

        avg_conf = sum(p["score"] for p in preds) / len(preds)

        # Log metrics
        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

        # Save and log predictions
        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })

        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(
            run_name,
            "-> accuracy:", accuracy,
            "latency:", round(latency, 3)
        )

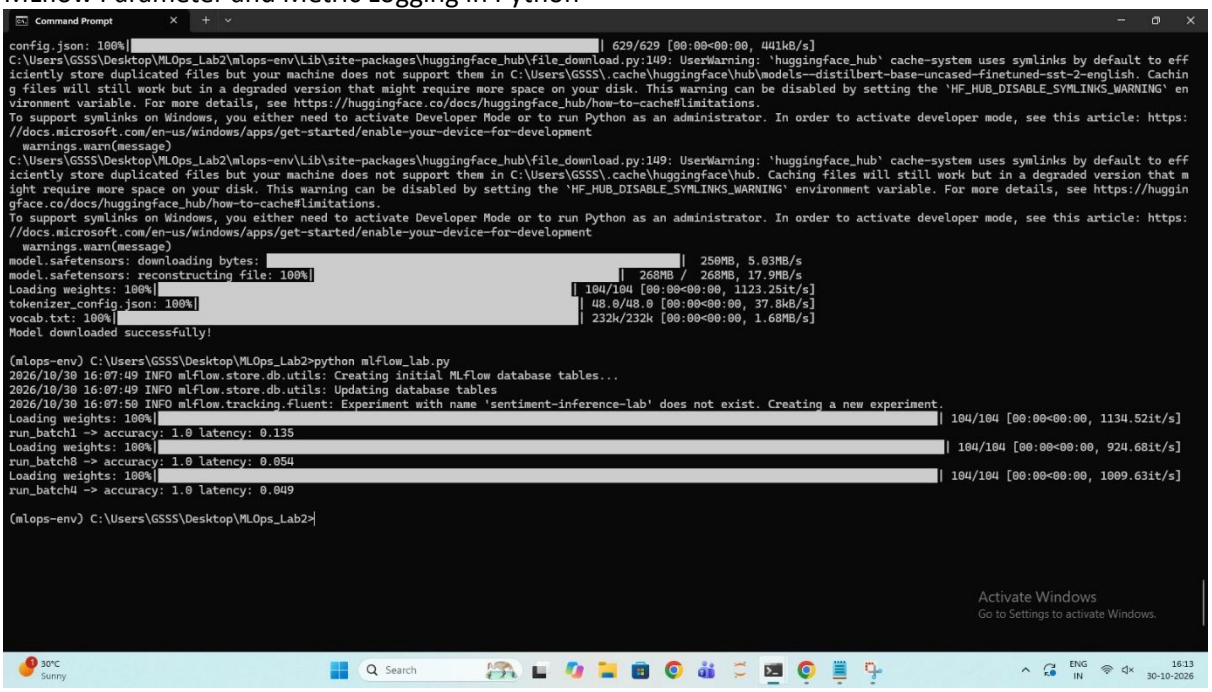

MODEL = "distilbert-base-uncased-finetuned-sst-2-english"

# Task 1: Two experiments
run_experiment(MODEL, batch_size=1, run_name="run_batch1")
run_experiment(MODEL, batch_size=8, run_name="run_batch8")

# Task 3: Third experiment
run_experiment(MODEL, batch_size=4, run_name="run_batch4")
```

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MLflow Parameter and Metric Logging in Python



Result:The sentiment analysis experiment was successfully executed using the DistilBERT pre-trained model and tracked using MLflow. Three experiments were performed with batch sizes 1, 4, and 8. All three runs achieved 100% accuracy. Among them, batch size 4 achieved the lowest latency of 0.049 seconds. MLflow successfully recorded and visualized the experiment parameters and performance metrics, making experiment comparison easier and more efficient.

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